

RESEARCH AREAS

Computer Vision, Generative Models, Anomaly Detection, Scene Understanding, Scene Graph Generation

EDUCATION

B.S. in Computer Engineering, *University of Tehran* Sep 2021 - Present

- Cumulative GPA: *A - 18.81/20*
- Last Two Years' GPA : *A - 19.38/20*

PUBLICATIONS

CELAD: Compositional Evaluation for Logical Anomaly Detection Under review for ICLR 2026
Ehsan Hassanbeygi, *Javad Kavian*, Kiarash Kiani Feriz, Mohammad Hossein Rohban

EXPERIENCE

Research Assistant, Sharif University of Technology Nov 2024 - Present
Robust and Interpretable Machine Learning Lab

- Supervised by [Professor Mohammad Hossein Rohban](#)
- Conducted research on Logical Anomaly Detection.
- Introduced an image dataset featuring compositional relations.
- Conducted benchmarking experiments by refining baseline implementations and integrating our evaluation pipeline, resulting in over 700 experimental runs.
- Proposed a novel method that achieved state-of-the-art results.
- Investigated methods from other fields such as scene graph generation, object detection, and segmentation to inspire new ideas.

Research Intern, University of Tehran May 2024 - Sep 2024
Advanced Robotics & Intelligent Systems Lab

- Supervised by [Professor Hadi Moradi](#)
- Researched machine learning approaches for ADHD diagnosis using behavioral data derived from cognitive game performance.
- Implemented and evaluated multiple ML and ensemble models, complemented by hypothesis testing to assess statistical significance.

RELEVANT COURSEWORK

Deep Generative Models (Graduate Level)

Grade: 20/20

Transcript name: Advanced Topics in Control 1

Machine Learning (Graduate Level)

Grade: 20/20

Artificial Intelligence

Grade: 20/20

Engineering Probability and Statistics

Grade: 19.3/20

Engineering Mathematics

Grade: 20/20

Computer Networks

Grade: 19.8/20

Top of class

Computer Aided Digital System

Grade: 20/20

Data Structures and Algorithms

Grade: 19.2/20

Operating System

Grade: 19/20

Introduction to Distributed computing

Grade: 20/20

TECHNICAL SKILLS

Machine Learning Tools: PyTorch, PytorchLightning, Hugging Face Transformers Library, Ultralytics

Programming Language & Tools: Python, C/C++, Java, R, MATLAB, Numpy, Pandas, scikit learn, Git, Linux, Latex

TEACHING EXPERIENCE

Deep Generative Models(Graduate Level)	1 semester
Designed and implemented an assignment on normalizing flows, including the development of a <i>Masked Autoregressive Flow (MAF)</i> model and conducting an anomaly detection experiment to evaluate its performance.	
Data Science	1 semester
- Designed and implemented an assignment on the <i>Langevin Dynamics</i> algorithm. - Created a tutorial video on sampling methods, including <i>Langevin Dynamics</i>, <i>Gibbs sampling</i>, and <i>rejection sampling</i>. - Designed and implemented an assignment on deep learning.	
Machine Learning (Graduate Level)	2 semesters
- Designed sample questions on <i>Karush–Kuhn–Tucker (KKT)</i> conditions. - Designed and implemented an assignment on the <i>Sequential Backward Selection (SBS)</i> model selection algorithm.	
Artificial Intelligence	3 semesters
- Designed and implemented assignments on the <i>SAMME</i> algorithm and <i>Decision Trees</i>. - Designed and implemented an assignment on <i>Deep Learning</i>.	
Programming Languages and Compilers	2 semesters
- Designed a new programming language for student projects and implemented its <i>name checker</i> and <i>type checker</i>. - Designed a written assignment on <i>type checking</i>.	
Introduction to Computing Systems and Programming	1 semester
Conducted project evaluations by interviewing students.	

PROJECTS

Deep Generative Models

- Implemented various generative models in **PyTorch** as part of the course assignments
- Models include **Diffusion Models** (DDPM and DDIM), **W-GAN**, **Normalizing Flow**, and **VAE**
- Developed a simulator for the **Annealed Langevin Dynamics** algorithm
- Conducted a **Visual Question Answering** experiment on the Paligemma VLM using clevr dataset

Transformer from Scratch

- Implemented all components of the Transformer architecture like multi-head self-attention, **multi-head cross attention**, **positional encoding**, **encoder**, and **decoder** layers from scratch using Pytorch, following the "Attention Is All You Need" paper.
- Preprocessed and tokenized bilingual text data to prepare it for training the translation model.
- Trained the model for translation task from english to persian

EEG Motor Imagery Analysis

- Preprocessed EEG data using Bandpass filtering, Common Average Referencing (CAR), Laplacian Filtering and Principal Component Analysis (PCA)
- Extracted features from EEG data using Independent Component Analysis (ICA) and Common Spatial Patterns (CSP)
- Classified signals using Logistic Regression, SVM, KNN, MLP and used ensemble methods like AdaBoost and XGBoost.

- Evaluate performance of models using Accuracy, Confusion Matrix and ROC.

Machine Learning Assignments

- Implemented various machine learning models from scratch as part of a course assignment.
- Covered topics including Bayes decision theory, KNN, Parzen window, Regression, Decision Trees, the SAMME algorithm, SVMs, Neural Networks, PCA, and GMMs.

Artificial Intelligence Assignments

- Completed AI projects on machine learning, clustering, deep learning, and reinforcement learning, including a Genetic Algorithm for Knapsack and HMM-based voice recognition.
- Deployed a CNN for tweet sentiment classification, performed clustering on the Iris dataset, and implemented a Q-Learning agent for a snake game.

Distributed Systems Projects

- Implemented core distributed systems concepts through course assignments.
- Covered topics including Raft consensus, concurrent database access under network failures, and MapReduce.

LANGUAGES

- **Persian:** Native
- **English:** IELTS Academic Band 8 (Listening: 8.5, Reading: 8.5, Writing: 7.0, Speaking: 7.0)

AWARDS & RANKINGS

- Ranked 83 among 126740 participants in the Nation-wide Iranian University Entrance Exam in Mathematics and Physics